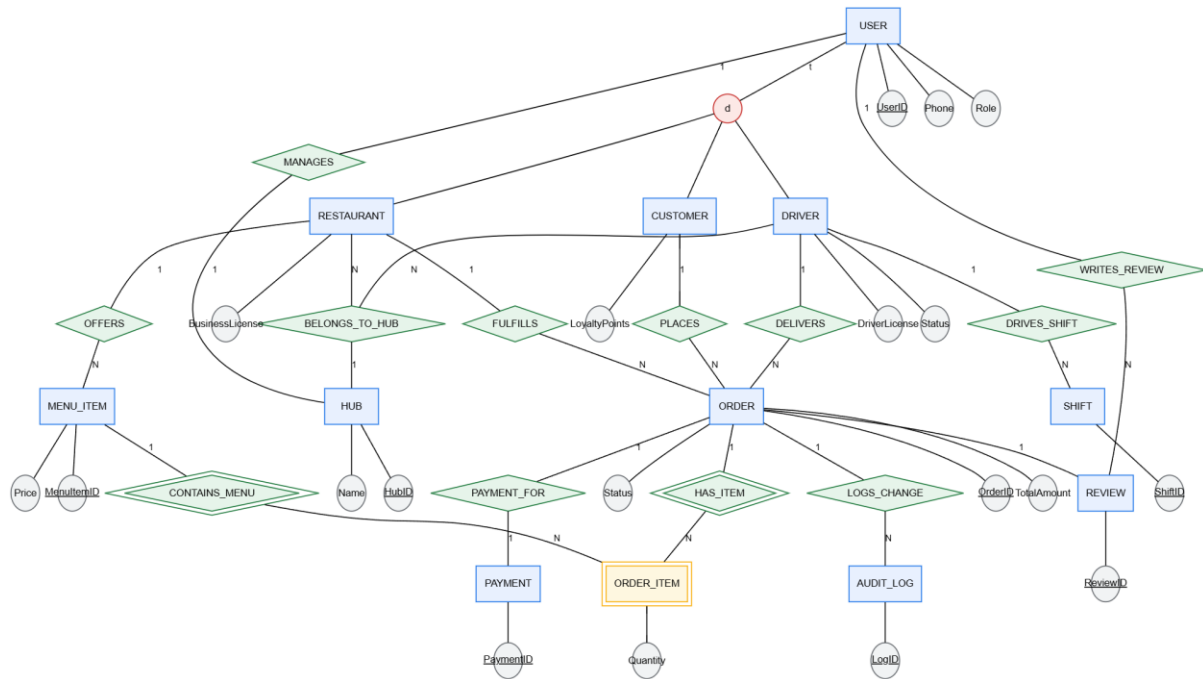




This design models a food delivery aggregator ecosystem based on the provided SQL schema (food_delivery_db):

- Superclass & Subclasses Hierarchy:** The superclass **USER** is specialized into three disjoint subclasses: **CUSTOMER**, **RESTAURANT**, and **DRIVER** (\$d\$) based on user roles. Each subclass inherits common authentication and profile fields (UserID, Name, Phone, Email, Password) while maintaining distinct attributes (e.g., LoyaltyPoints for Customer, BusinessLicense for Restaurant, and DriverLicense for Driver).
- Hub Management:** The **HUB** entity handles regional distribution and is managed by a **USER** (ManagerID), while grouping local **RESTAURANT** and **DRIVER** entities.
- Order Processing & Menu Management:** Each **RESTAURANT** offers multiple **MENUTITEM** entities. An **ORDER** acts as a transaction hub connecting a **CUSTOMER**, a **RESTAURANT**, and an assigned **DRIVER**. The **ORDERITEM** weak entity captures specific menu choices and quantities for each order.
- Supporting Services:** **PAYMENT** processes financial transactions for orders; **SHIFT** tracks driver working hours and bonuses; **REVIEW** stores user feedback; and **AUDITLOG** tracks historical order status transitions for system auditing.